10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 10.2 — Construction of Tangents

Unit 10 | Solved Exercise | New Book (PECTAA)

Exercise 10.2 (Solved)

Every tangent here is drawn the same way: the radius at the point of contact must be perpendicular to the tangent. So first find the radius, then draw a line at 90° to it at the point of contact.

1. Draw tangent to arc APB at point P, when P is midpoint of the arc.

1. Join the end points A and B of the arc.
2. Draw the right bisector LM of the chord AB.
3. This right bisector passes through P, the midpoint of the arc, and also through the centre O of the circle.
4. At P draw a line perpendicular to OP.
5. This perpendicular line is the required tangent at P.

Answer: Tangent drawn at the midpoint P of the arc

2. Draw tangent to arc PQR at point P, when P is endpoint of the arc.

1. Take any point A on the arc and join P to A.
2. Draw right angle PAN so that AN cuts the arc at B.
3. Join B to P.
4. Construct m∠APR = m∠ABP, using the compasses.
5. Produce RP towards P. Then RPQ is the required tangent.

Answer: Tangent RPQ drawn at the end point P of the arc

3. Draw tangent to arc ABC from a point P, when P is outside the arc.

1. Join A to P; AP cuts the arc at Q.
2. Find the midpoint M of AP.
3. Take AMP as diameter and draw a semi-circle.
4. Draw perpendicular QS on AP, which cuts the semi-circle at R.
5. Take mRP as radius and draw an arc that cuts the arc ABC at D.
6. Join P to D and produce towards D; PD is the required tangent at D.

Answer: Tangent PD drawn to the arc from the outside point P

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4. Draw a circle of radius 1.3 cm and draw a tangent at point P, when P lies on its circumference.

1. Take O as centre and radius 1.3 cm, and draw the circle.
2. Mark the point P on the circumference.
3. Join O to P. OP is the radius at the point of contact.
4. At P construct a right angle on OP with the help of compasses.
5. Produce the perpendicular on both sides of P; this line is the required tangent.

Answer: Circle of radius 1.3 cm drawn with the tangent at P

5. Draw a circle of radius 1.5 cm and draw a tangent at point P, when P is at a distance of 8 cm from its centre.

1. Take O as centre and radius 1.5 cm, and draw the circle.
2. Produce the radius outwards and mark a point P on it such that OP = 8 cm (P lies outside the circle).
3. Join O to P and find its midpoint M.
4. Take M as centre and radius mOM = mMP, and draw a semi-circle. It cuts the given circle at T.
5. Join P to T and produce it; PT is the required tangent.
6. The other tangent can be drawn by taking the second point where the semi-circle cuts the circle.

Answer: Two tangents drawn from the outside point P, each of the required length

6. Draw a circle of radius 1.6 cm. Draw two tangents that meet an angle of 30°.

1. Draw a circle of radius 1.6 cm with centre O and draw a diameter AOB.
2. At A draw a perpendicular to OA; this is the first tangent.
3. At O draw an angle of 30°, so that m∠AOC = 30°, with C on the circle.
4. At C draw a perpendicular to OC; it cuts the first tangent at the point P.
5. PA and PC are the two required tangents, meeting at m∠APC = 30°.

Answer: Two tangents drawn meeting each other at 30°

7. Draw a tangent to any point on a circular part of the track having radius = 2 cm.

1. Draw the circular track with radius 2 cm and centre O.
2. Take any point P on the circular track.
3. Join O to P; this is the radius at P.
4. At P draw a perpendicular to OP.
5. This perpendicular line is the required tangent to the track at P.

Answer: Tangent drawn at a point on the circular track

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8. From a pulley point, construct two tangents to a machine wheel having r = 2.1 cm, such that the angle between them is 30°.

1. Draw the machine wheel with centre O and radius 2.1 cm, and draw a diameter AOB.
2. At A draw a perpendicular to OA; this is the first tangent.
3. At O draw m∠AOC = 30°, taking C on the circle of the wheel.
4. At C draw a perpendicular to OC; it cuts the first tangent at the point P.
5. PA and PC are the two required tangents, and the angle between them at P is 30°.

Answer: Two tangents drawn to the wheel enclosing 30°

Remember — construction of a tangent

Point on the circle: join the centre to the point, then draw a perpendicular there.

Point outside the circle: take the mid-point of OP, draw a semi-circle on OP, join the cutting point to P.

Two tangents enclosing an angle θ: draw a diameter, draw one tangent at its end, and take the angle θ at the centre.

A tangent is always perpendicular to the radius at the point of contact.

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